

Nathan Charles Lopes

Software Developer | Cybersecurity & ML Background

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About

IT graduate student at the University of Melbourne with experience in Python, web development, and cybersecurity tools including Wireshark, Nmap, and Metasploit. Built a full-stack SIEM platform combining detection engineering with AI-driven alert investigation. Interested in applying software development and security skills through an IT internship

Skills

Programming & Web Development Python, Java, C, JavaScript, Django, Flask, FastAPI, HTML, CSS	Cybersecurity Tools Kali Linux, Nmap, Wireshark, Metasploit, Burp Suite, Nessus
Databases Firebase, PostgreSQL	Cloud Platforms AWS
Tools & Platforms Git, GitHub, Linux, Docker	Soft Skills Leadership, Communication, Team Collaboration, Time Management, Critical Thinking

Work experience

IT support, James Battery (Part Time) 	06/2022 – 01/2025 Mumbai
- Supported end-to-end IT operations, including website development and maintenance for the company. - Set up and configured office systems, including computers, routers, and internet connectivity. - Assisted with database/data entry tasks and digital record management. - Helped implement basic security infrastructure, including CCTV setup and monitoring support. - Provided general IT troubleshooting and technical support across office systems.	

Technical Projects

Custom SIEM - Security Information & Event Management Platform , <i>(Python, FastAPI, PostgreSQL, Docker, LLM Agents)</i>	
- Built a decoupled SIEM backend detecting 7+ attack patterns (brute force, cross-host attacks, web scanning) mapped to MITRE ATT&CK. - Designed an AI agent using locally hosted LLMs to autonomously investigate alerts via tool calls with auditable reasoning. - Implemented CI/CD, 30+ automated tests, and secret scanning for a 5-service Docker architecture.	
TrueMatch - Behaviour-Based Matching System , <i>(Python, Firebase, Android)</i>	
- Built a Python backend that processes behavioural data from Firebase Firestore to generate compatibility scores. - Designed pipelines analysing smartphone sensor data (steps, location, app usage, sleep). - Integrated backend services with Android clients for real-time synchronization.	
Edenthought - Personal Journal Web App , <i>(Django, Python, SQLite, Bootstrap 5)</i>	
- Built a full-stack web app enabling user registration, login/logout, profile management, and CRUD operations for journal entries. - Implemented email notifications on registration and secure dashboards with authentication controls. - Developed responsive UI with Bootstrap 5 and managed static/media files efficiently.	
Stock Market price prediction using Stacked LSTM , <i>(ML, LSTM)</i>	
- Developed a stacked LSTM neural network for stock price trend prediction. - Implemented data preprocessing, feature scaling, and time-series visualization. - Achieved ~70% trend prediction accuracy on historical stock data.	

Education

Master of Information Technology (Specialisation Cybersecurity) , <i>The University of Melbourne</i> WAM: 72.83/100	11/2026 Melbourne, VIC
Bachelor of Computer Engineering with Honors in Cyber Security , <i>Fr. Conceicao Rodrigues College of Engineering (Mumbai University)</i> CGPA: 9.36/10	07/2024 Mumbai

Certifications

Google Cybersecurity Certificate 04/2025 Coursera	Complete Ethical Hacking Bootcamp 09/2024 Udemy	Python Django: Ultimate Beginners Course - 2024 03/2024 Udemy
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Volunteer Experience

National Service Scheme - CRCE, Volunteer	07/2022 – 07/2024 Mumbai
- Organised campus cleanliness drives and blood donation drives. - Taught underprivileged children and led a 5-member student team.	